

CLOUD REDUNDANCY IN CLOUD COMPUTING

RELIABILITY • AVAILABILITY • DISASTER RECOVERY

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INTRODUCTION TO CLOUD REDUNDANCY

- Cloud computing provides scalability and agility.
 - Risks include outages and cyber threats.
 - Redundancy ensures resilience and business continuity.

DEFINITION OF CLOUD REDUNDANCY

- Multiple copies of servers, data, networks, applications.
 - Automatic failover during failure.
 - Minimizes user disruption.

IMPORTANCE FOR BUSINESSES

- Prevents revenue loss due to downtime.
 - Protects intellectual property.
 - Improves customer trust.

TYPES OF REDUNDANCY OVERVIEW

- Data redundancy
 - Network redundancy
 - Application redundancy
 - Component redundancy
 - Geographic redundancy

DATA REDUNDANCY

- Multiple data copies across locations.
 - Protection against corruption and ransomware.
 - Improves availability.

NETWORK REDUNDANCY

- Backup routers and switches.
 - Multiple internet connections.
 - Reduces connectivity failures.

APPLICATION & COMPONENT REDUNDANCY

- Multiple app instances for load balancing.
 - Extra servers and storage devices.
 - Ensures high availability.

GEOGRAPHIC REDUNDANCY

- Replication across regions.
 - Disaster recovery capability.
 - Global service continuity.

HOW REDUNDANCY WORKS

- Replication across regions.
 - Redundant infrastructure.
 - Failover & load balancing.

BENEFITS OF REDUNDANCY

- High availability & uptime.
 - Data protection.
 - Improved scalability.
 - SLA compliance.

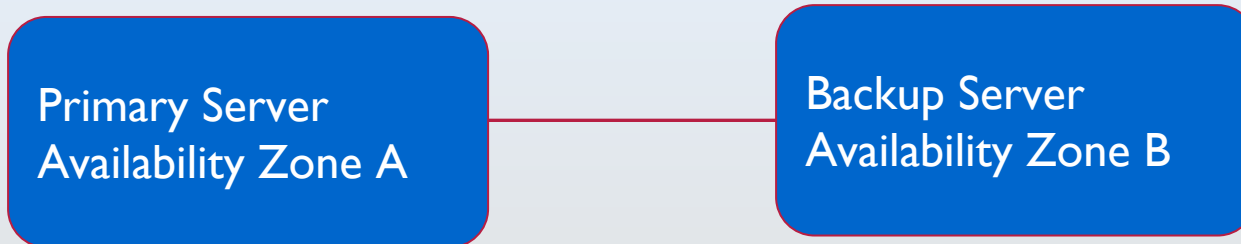
CHALLENGES OF REDUNDANCY

- Higher cost.
 - Complex implementation.
 - Testing and validation needs.

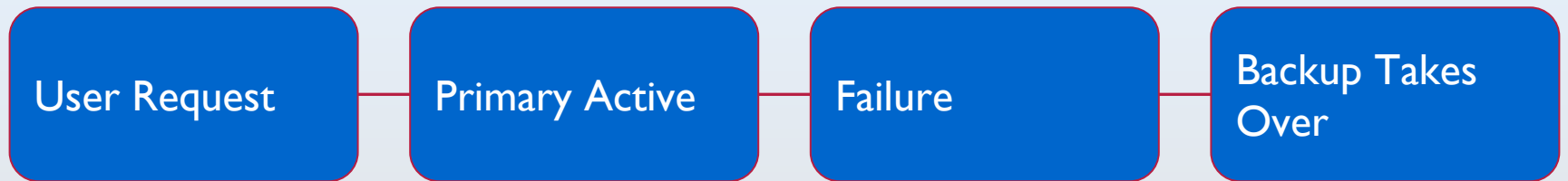
BEST PRACTICES

- Plan redundancy strategy.
 - Adopt multi-cloud approach.
 - Continuous monitoring.
 - Automated failover testing.

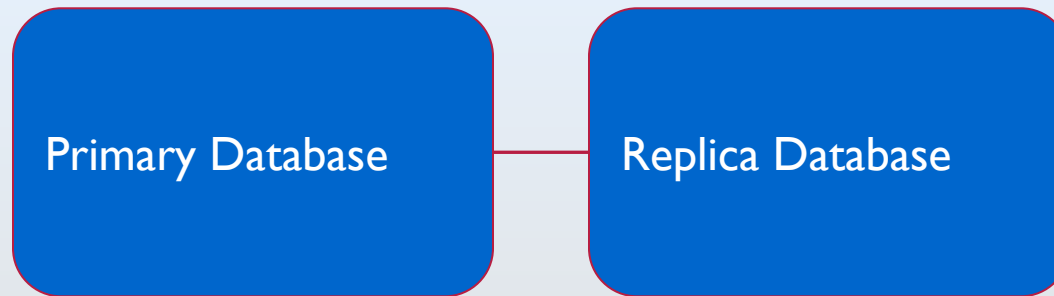
REDUNDANT CLOUD ARCHITECTURE



AUTOMATIC FAILOVER PROCESS



DATA REPLICATION MODEL



MAJOR CLOUD PROVIDERS

AWS

Microsoft
Azure

Google
Cloud

Akamai

Failover System in Cloud

FAILOVER SYSTEM IN CLOUD

The ~~failover system technique~~ makes use of established clustering technologies to provide redundant implementations in order to boost the availability and reliability of IT resources. When an active IT resource becomes unavailable, a failover system is set up to automatically transition to a redundant or standby instance of that resource.



CONFIGURATIONS OF FAILOVER SYSTEMS

In an active-active setup, redundant IT resource implementations actively and synchronously support the workload (Figure 1). It is necessary to load balance among active instances. The failed instance is eliminated from the load balancing scheduler when a failure is discovered (Figure 2). Whenever a fault is discovered, the processing is transferred to the IT resource that is still active (Figure 3).

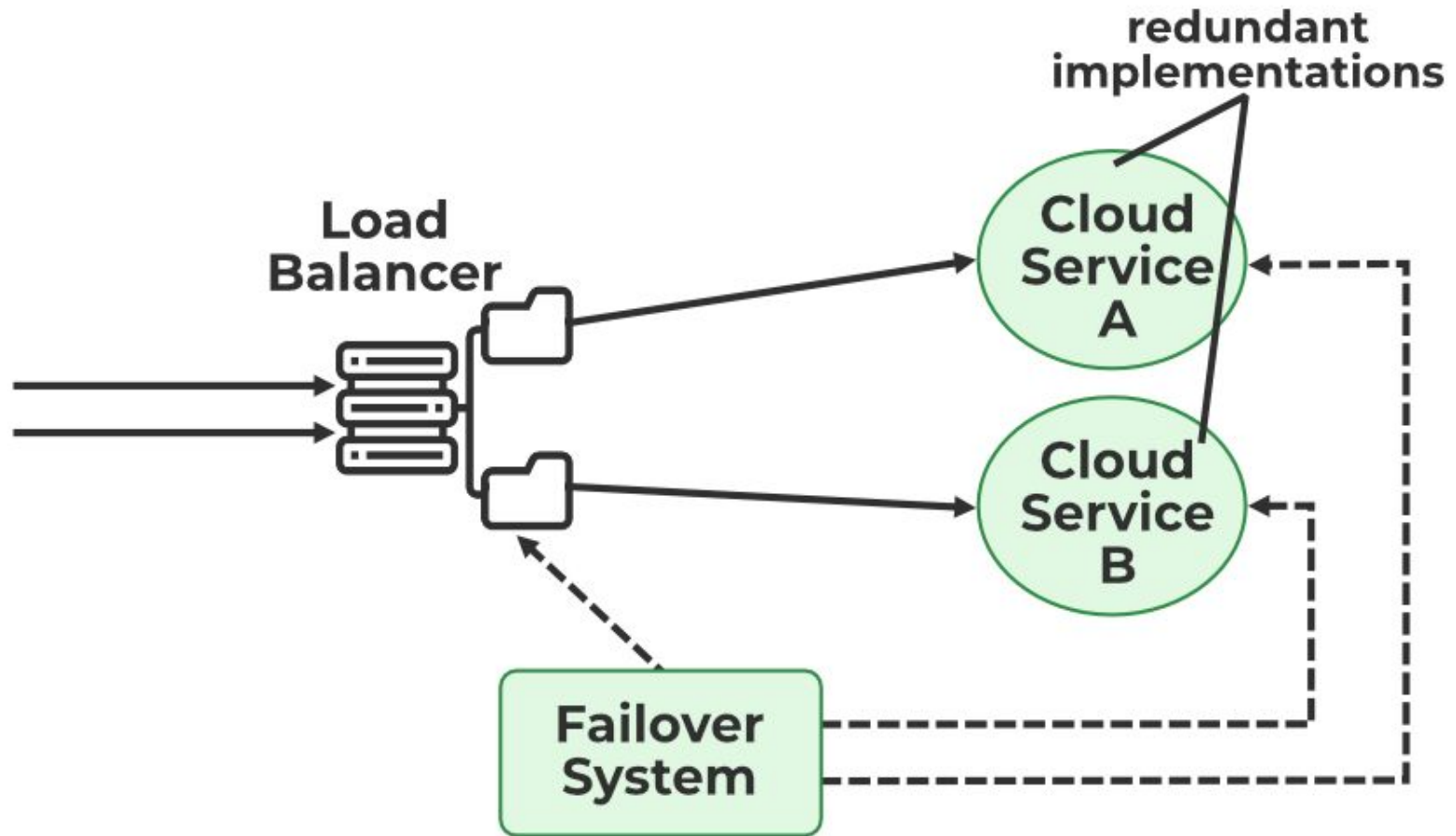


Figure 1: shows how the failover system monitors Cloud Service A and B operational conditions.

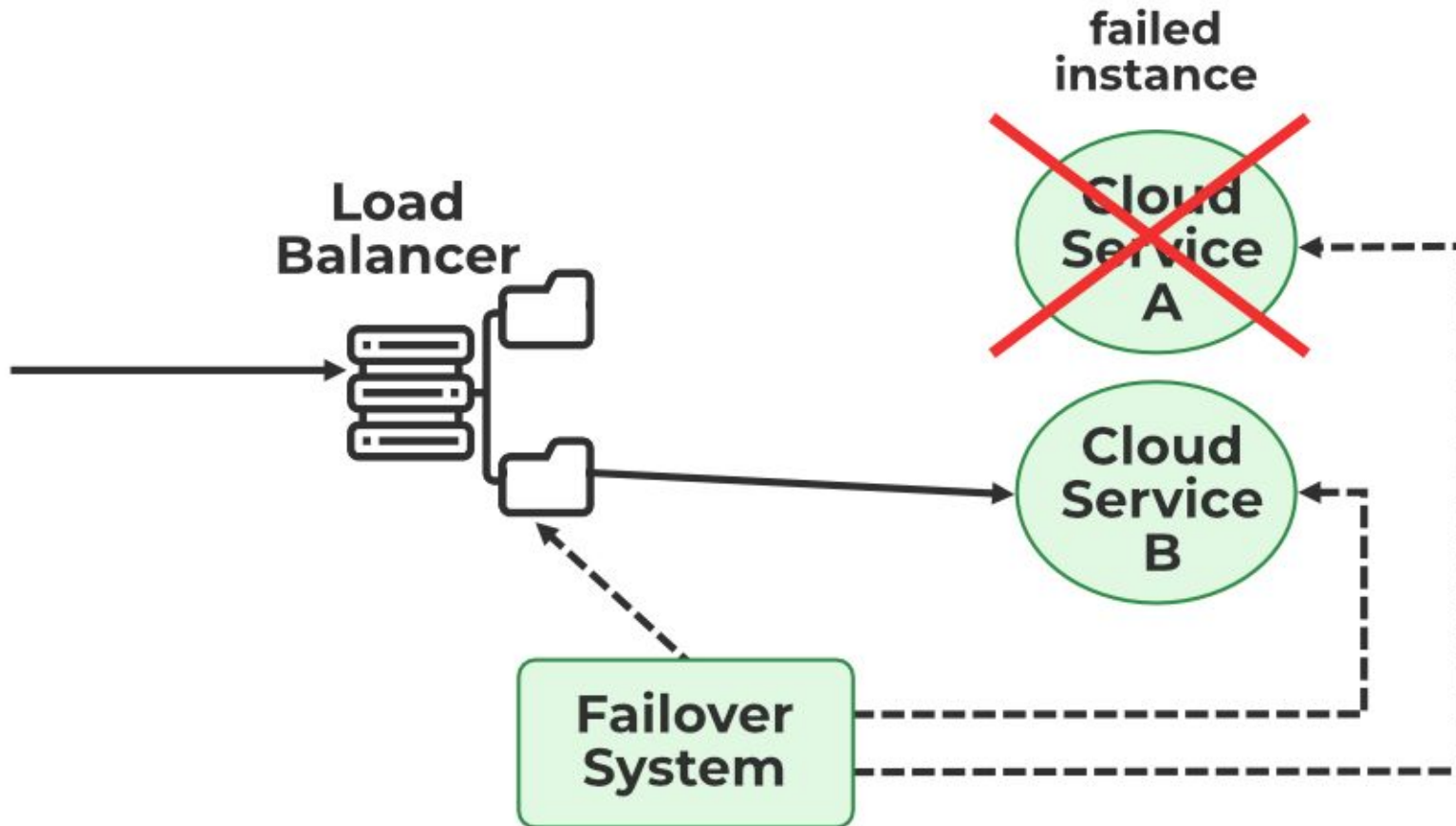


Figure 2: The failover system instructs the load balancer to transfer the workload to the redundant Cloud Service B implementation when a failure in one Cloud Service A implementation is discovered.

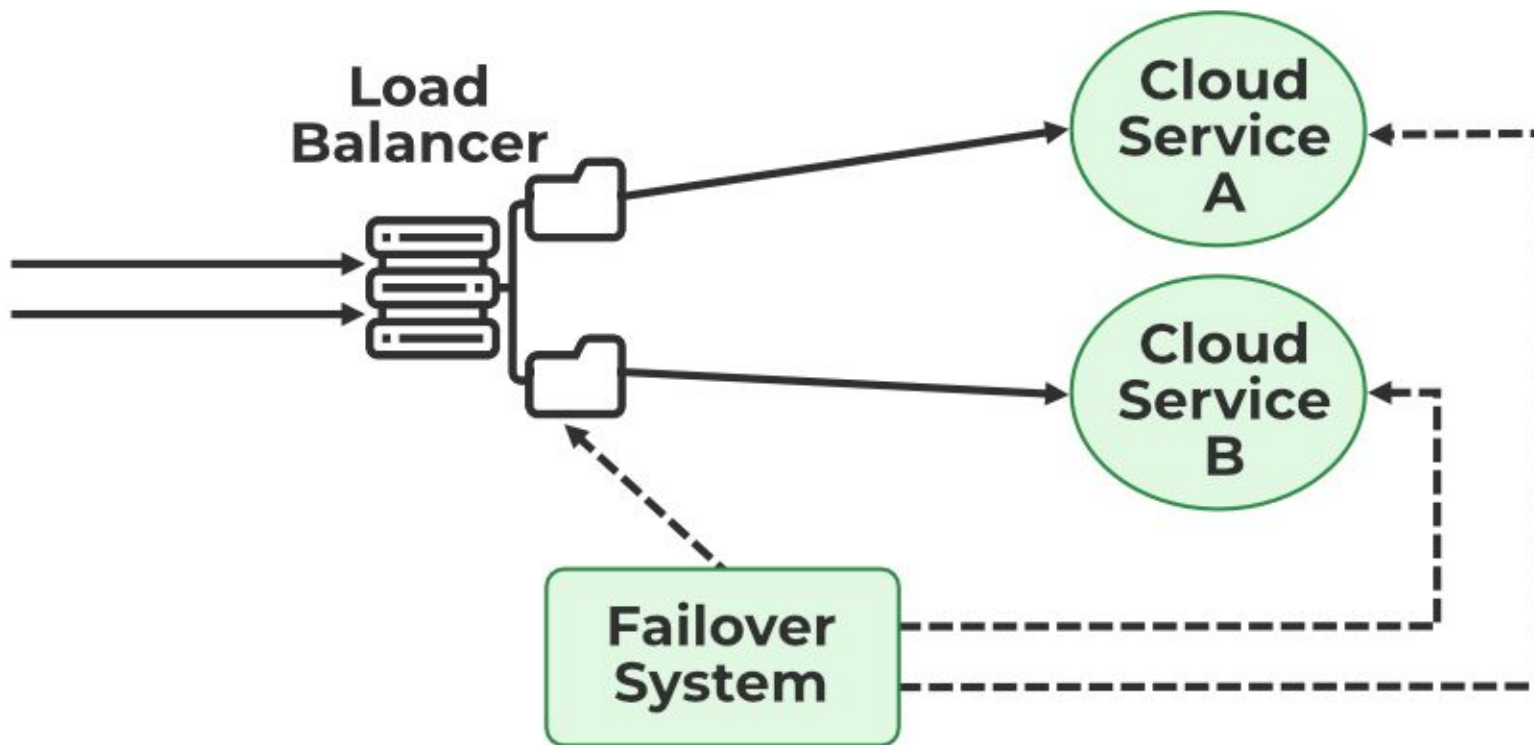


Figure 3: Failure of the Cloud Service After being duplicated from or recovered from an implementation, a cloud service is operational. The failover system has now sent the load balancer instructions to disperse the load.

ACTIVE-PASSIVE

A standby or inactive implementation is triggered in an active-passive setup to take over processing from an IT resource that becomes unavailable, and the associated workload is directed to the instance taking over the operation.

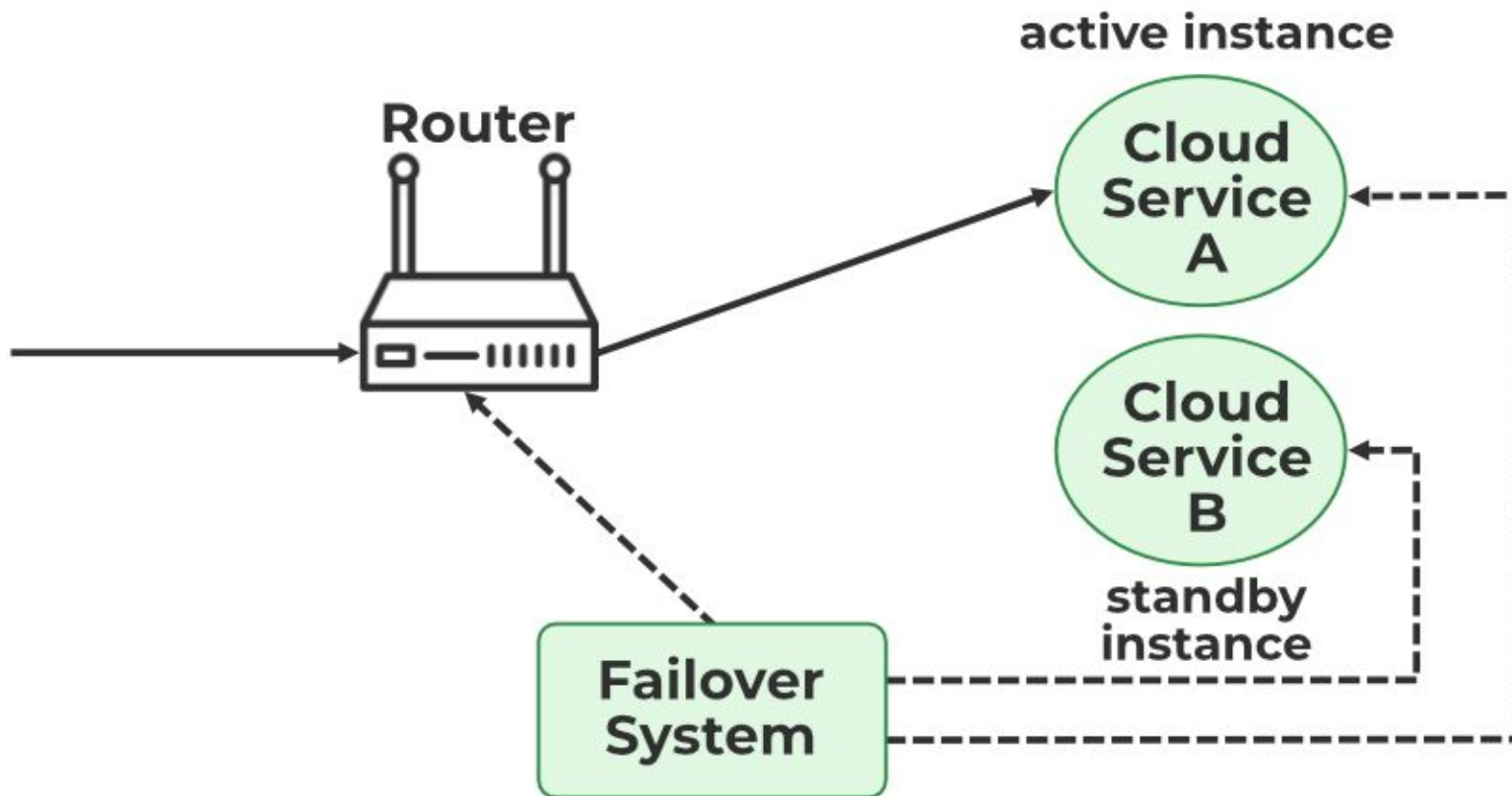


Figure 4: The failover system keeps track of Cloud Service's operational condition. In the Cloud Service, first Requests from cloud service users are being processed by an implementation serving as the active instance.

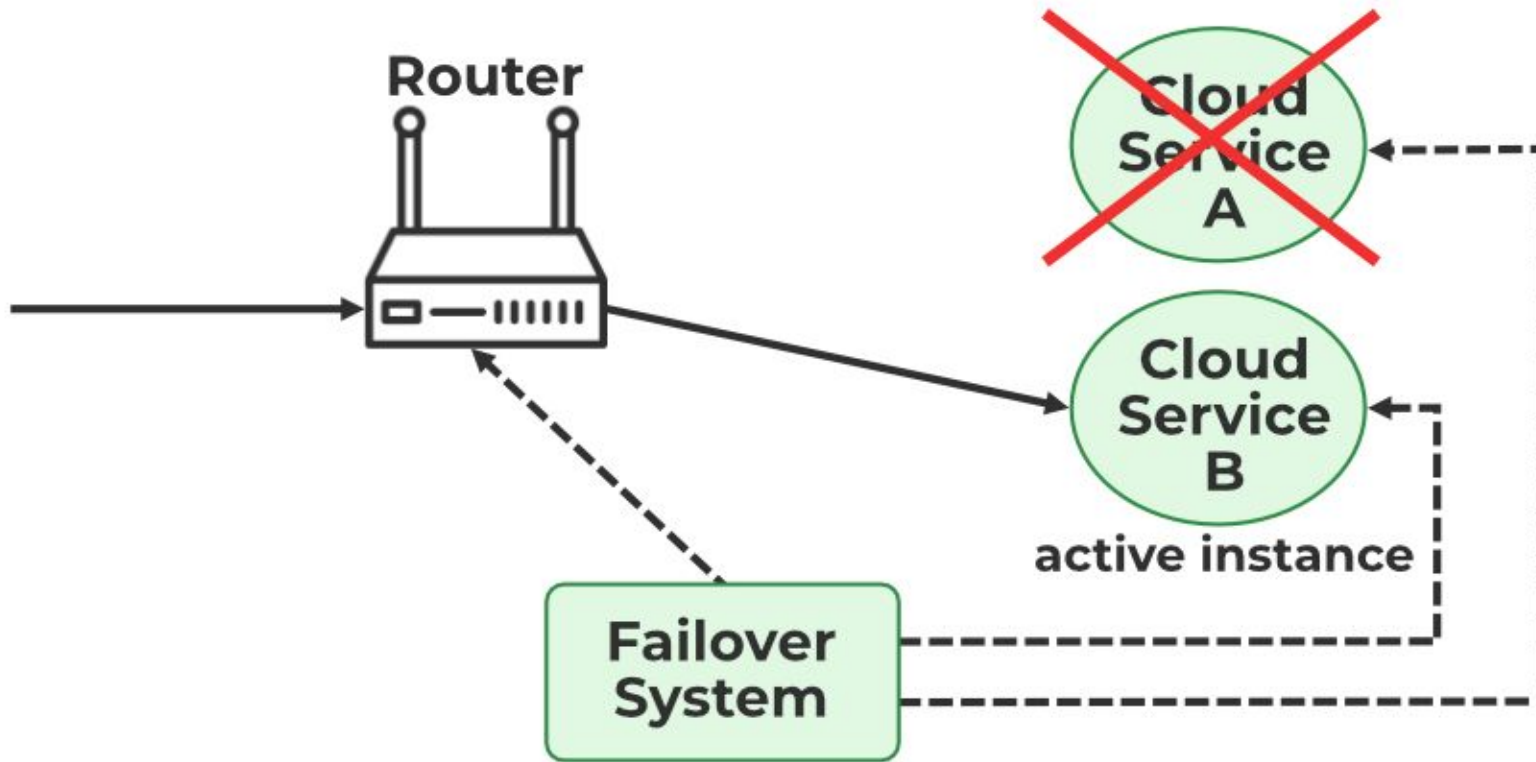


Figure 5: Service in the Cloud when an implementation serving as the active instance experiences a failure, the failover system notices it and switches to using the inactive Cloud Service. An implementation is made, and the workload is directed there. The most recent Cloud Service invoked The active instance has now been replaced with an implementation.

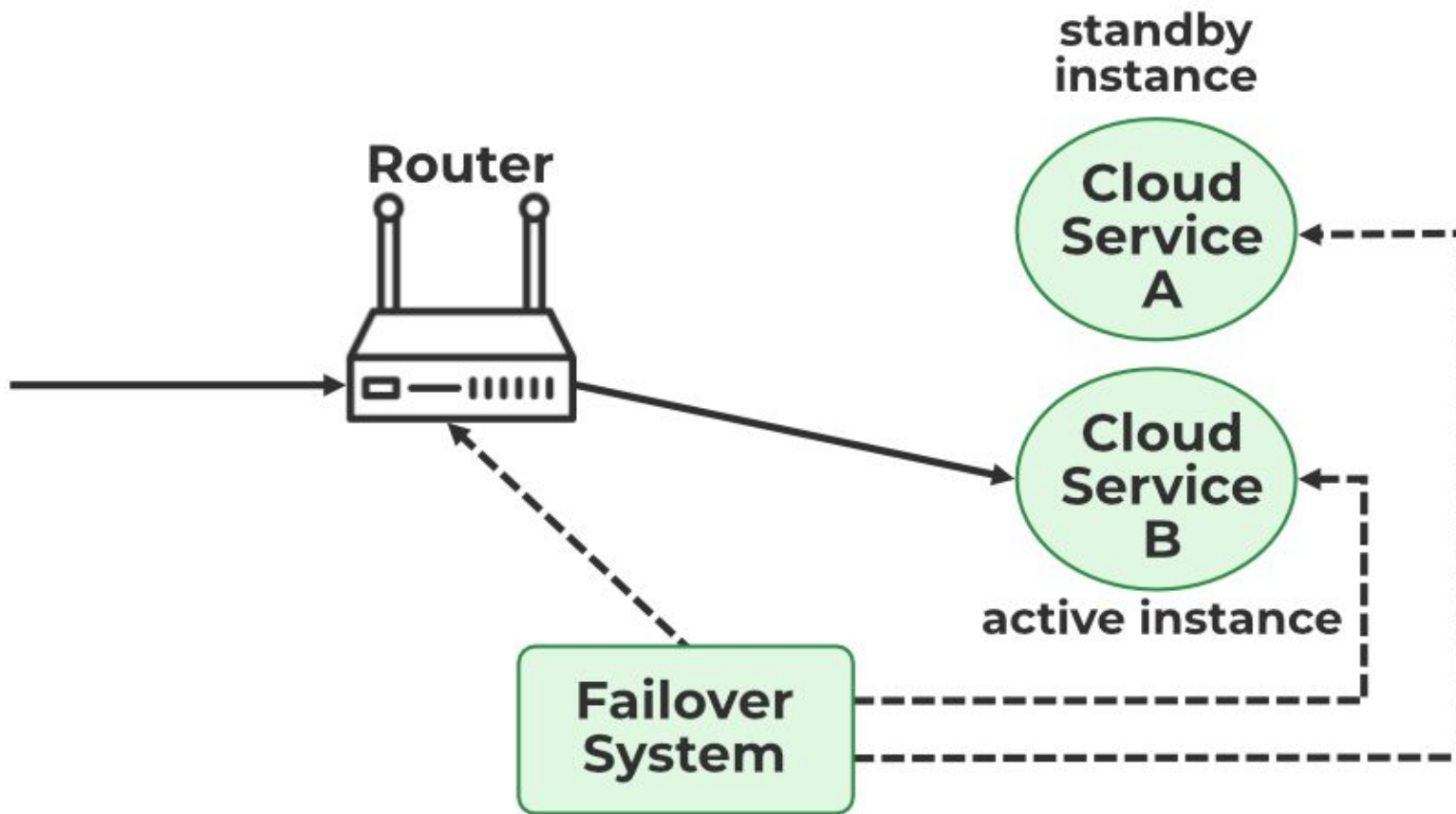


Figure 6: The unsuccessful Cloud Service A implementation is salvaged or copied into a working cloud service and is now designated as the standby instance, with the first invoked Cloud Service A continuing to function as the active instance.